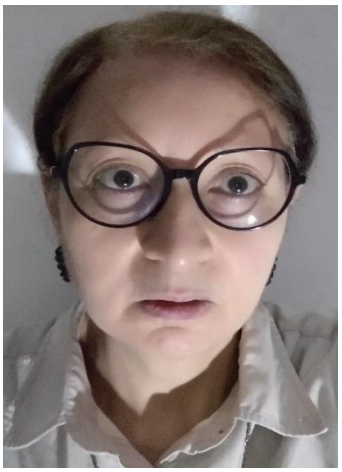
Resume



Personal

Name ***Shymaa M. Arafat***

Birth Date 25-7-1972

Birth Place *Malaysia*

Email Shymaa.arafat@gmail.com, shar.academic@gmail.com

Phone 01002033426

GitHub https://github.com/DrShymaa2022/,
https://github.com/Shymaa-Arafat/

Google Scholar https://scholar.google.com/citations?hl=en&user=ivrUdmYAAAAJ

Medium https://medium.com/@shymaa.arafat/

ORCID iD: 0009-0009-7571-4060

Facebook https://m.facebook.com/100010333725264/

YouTube. https://youtube.com/@shymaaarafat1342

Researchgate https://www.researchgate.net/profile/Shymaa-Arafat

Wallet address 0xe8a9e41b726ba3eae6837dcf74e00e2d224d5842

Address

Egypt. (but willing to travel and welcomes abroad positions)

Previous Positions

- Visting Associate Professor at Beirut Arab University, Lebanon (2/2012-4/2012), Faculty of Engineering, Computer Science & Engineering Dept.
- Part time Associate Professor in Computer Engineering Dept., Pharos University, Alexandria, Egypt (Spring 2011, Fall 2009)
- Full time Associate Professor at the Computer Science Dept., Faculty of Computer & Information Sciences, Ain Shams University, Cairo, Egypt. (started 1/10/1999 as an assistant lecturer, and resigned 7/7/2008)
- Participated as part-time lecturer/ professor in many universities & Institutes in the same Interval (99-2008) as listed in the teaching experience
- Research Assistant at Mubarak City for Scientific Research & Technological Applications, Informatics Institute (4/1995- 30/9/1999)

(worked mostly in a virtual reality application where we took a training of the *Worldup* Software, and worked on developing a walk thru application)

Education Record (Academic Grades)

Ph.D. in Computer Science (July 2001)
Alexandria University, Faculty of Engineering, Computer Science & System Engineering Dept.
https://srv3.eulc.edu.eg/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=PublicDrawThesis&BibID=10011669
http://193.227.0.100/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=ThesisPicBody&BibID=10011669
(copy of the thesis is available at https://github.com/DrShymaa2022/PhD_papers)
Title of Thesis *Operating on Optimal Binary Code Trees*
Supervisors
Prof. Dr. Ahmed A. Belal
Prof. Dr. Mohamed S. Selim

Ph.D. Courses

Course Name	Teaching Professor	Grade
Parallel & Distributed Systems	Dr. Hussain Ali	V. Good
Security in Computer Networks	Dr. Ahmed Belal	Distinction
Probabilistic Algorithms	Dr. Ahmed Belal	Distinction

M.Sc. in Computer Science (July 1997)
University of Alexandria, Faculty of Engineering, Computer science & System Engineering Dept.
https://srv3.eulc.edu.eg/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=PublicDrawThesis&BibID=10153838
http://193.227.0.100/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=PublicDrawThesis&BibID=10153838
Title of Thesis *An Efficient Parallel Sample Sorting Algorithm*
Supervisors
Prof. Dr. Ahmed A. Belal
Prof. Dr. Nagwa M. el-Makky

Master Courses

Course Name	Teaching Professor	Grade
Networking Personal Computers	Dr. Walid Aref	Distinction
Advanced Studies on Data Bases	Dr. Nagwa el Makky	V. Good
Algorithms Design & Analysis	Dr. Ahmed Belal	Pass *
Modern Control Theory	Dr. Ibrahim Abd el Salam	Distinction
Hard Ware	Dr. Salah Selim & Dr. Layla Abu Hadid	V. Good
Natural Language Processing	Dr. Ahmed el Nahas	Pass

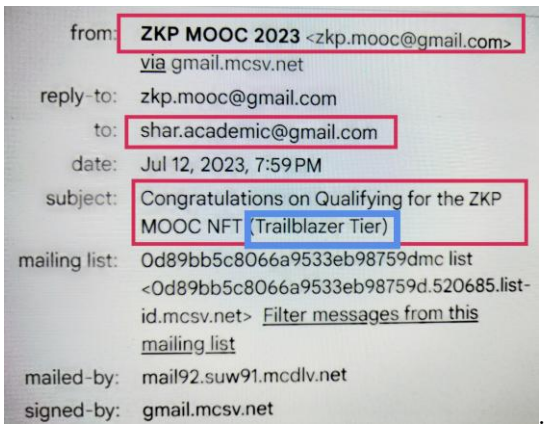
* The course is offered on a pass/Fail basis

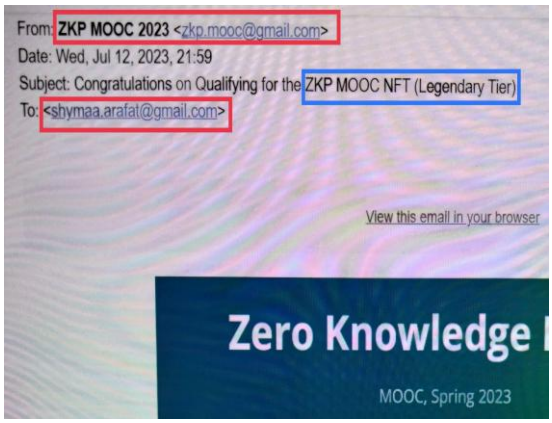
B.Sc. in Computer science & Automatic Control (June 1994)
University of Alexandria, Faculty of Engineering, Computer science & Automatic Control Dept.
Graduation Grade Very Good (84.39%) with degree of honor
Graduation Project Simulation of Storage Subsystems
Project Grade Distinction

Online MOOCs

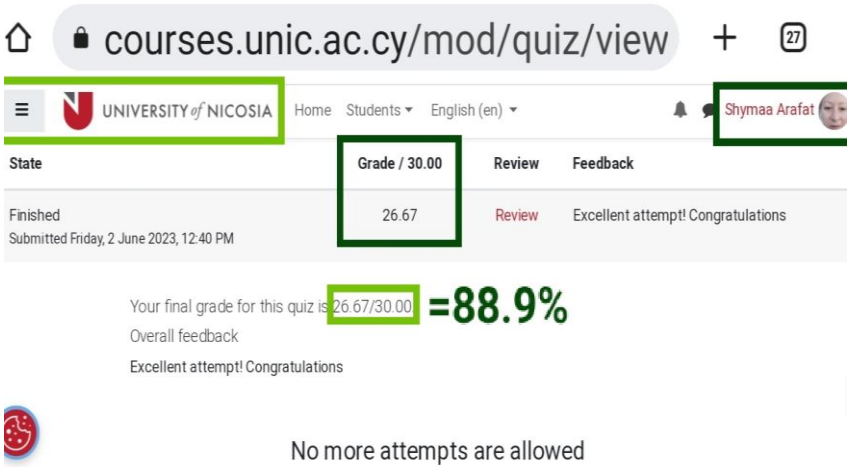
Mastering Web3 MOOC online course (winter 2025)
offered by University of Nicosia
Final grade: 100 (50/50 MCQ, download to view in detail
https://github.com/DrShymaa2022/recommendation_letters_CVdocuments/blob/main/Mastering_Web3_Final%20MOOC%20Quiz_%20Attempt%20review%20_%20unic.mhtml)
Certificate
https://github.com/DrShymaa2022/recommendation_letters_CVdocuments/blob/main/Mastering_Web3_Certificate.pdf
Verifiable on the Bitcoin blockchain (https://www.unic.ac.cy/verify/)

ZKP MOOC online course (Winter 2023)
Zero Knowledge course offered by Berkeley University
https://zkp-learning.org
Earned *Legendary Tier* (according to answering lecture quizzes on time+ delivering training lab + take home exam) & *Trailblazer Tier* (according to articles available on GitHub and medium) NFTs.
https://basescan.org/tx/0xb6e86833b0f796c9a641dcc50ad0a400ce0ad7c5210abfe6dc19fbb5559d92bd
http://basescan.org/tx/0x1f45129107b46fdea040761b6055e7a7f55326b0b40a72d0dbee536910523fdb


from: ZKP MOOC 2023 <zkp.mooc@gmail.com> via gmail.mcsv.net
reply-to: zkp.mooc@gmail.com
to: shar.academic@gmail.com
date: Jul 12, 2023, 7:59 PM
subject: Congratulations on Qualifying for the ZKP MOOC NFT (Trailblazer Tier)
mailing list: 0d89bb5c8066a9533eb98759dmc list <0d89bb5c8066a9533eb98759d.520685.list-id.mcsv.net> Filter messages from this mailing list
mailed-by: mail92.suw91.mcdlv.net
signed-by: gmail.mcsv.net

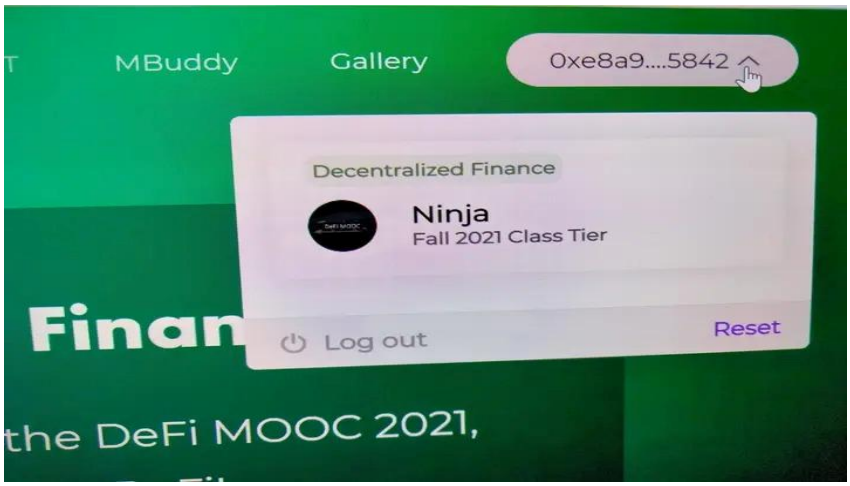


Introduction to DeFi MOOC (Spring 2023)
A compact 6 week online course offered by University of Nicosia
Final grade **88.9**, (26.67/30, No free NFT)
https://github.com/DrShymaa2022/recommendation_letters_CVdocuments/blob/main/DeF23_FinalQuiz_%20Attempt%20review%20_%20unic.mhtml (download to view)



<u>Introduction to Digital Currencies MOOC</u> (Spring 2023)
Offered by University of Nicosia
Final grade **85.67**, NFT issued on 23/5/2023.
https://app.block.co/certificate/f640aada-8353-43bb-9126-f6a64b829b90/

<u>DeFi MOOC online course</u> (Fall 2021)
Decentralized Finance online MOOC course offered by Berkeley University
https://defi-learning.org/f21
Earned the NFT *<u>Ninja-Tier</u>*
https://explorer.oasis.io/mainnet/emerald/token/oasis1qrels9582m4wf25guu5057wtg74jncvqay8u6947/instance/41
(The older site https://gallery.metamirror.space/detail/?item=defimooc2021nft-41 has changed due to changes in the oasis blockchain holding those NFTs that were somewhat earlier pioneers for such an idea of using NFTs as course certificates; passing was according to <u>MCQ</u> exam+an <u>article</u> that is available on my GitHub+<u>smart contract</u> lab exercise also available on GitHub).
-For honesty, I passed without getting the smart contract right then I've got it correct afterwards even with different versions (trying the 2 steps liquidation for > 50% that was not required by the lab, only presented in the lectures as a new contribution from the professor's paper).

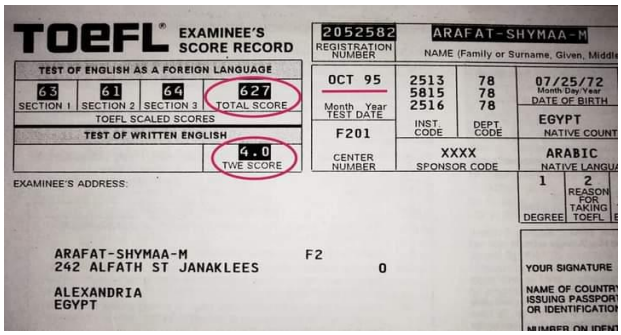


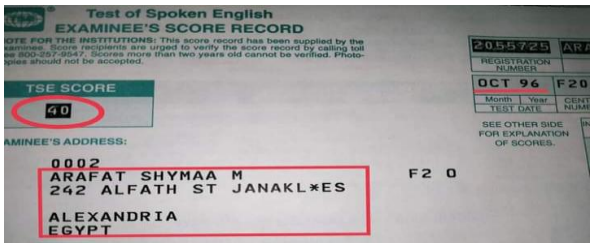
AMID-EAST Exams

TOEFEL **627** in 10/1995

TWE **4** in 10/1995

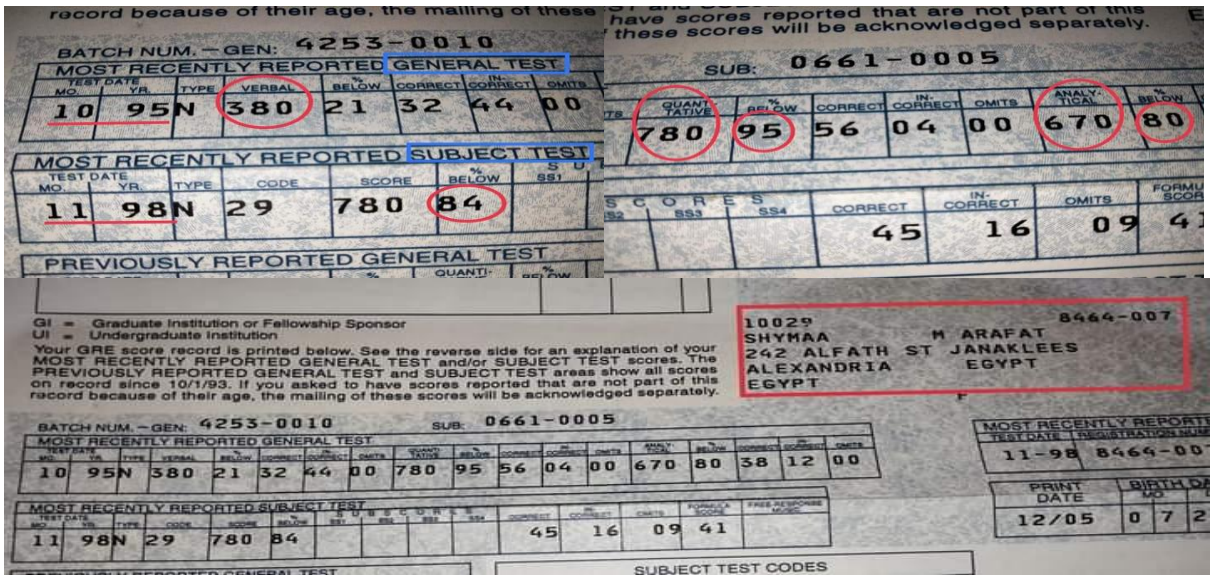
TSE **40** in 10/1996





GRE General
verbal **380**, **analytical** **670** , **quantitative** **780** in 10/1995

GRE Subjective in Computer Science **84** in 11/1998



Activities

Winner of a research grant offered by the Center of Special Studies and Programs (CSSP), Bibliotheca Alexandrina (March 2005,involved 2 weeks visit to SMU11/9/2005-25/9/2005)
https://www.bibalex.org/cssp/Researchs/2004.htm
https://www.bibalex.org/cssp/Event/Event.aspx?ID=55
Research Title: *Providing Secure Communications for Wireless and Wireline Applications*
Foreigner Counterpart: Prof. Dr. Hesham el Rewini,
Chairman of Computer Science & Engineering Dept., Faculty of Engineering, Southern Methodist University (SMU), USA (now in Marymount University as stated in the recommendation letter)

Teaching Experience

Course Name	Academic year	University	Date
Algorithms Design & Analysis	Post-graduate Computer sciences	Faculty of Science, Alexandria University	3/2004-/2004 3/2003-/2003
Compilers Theory	4th year Computer science		9/2003-1/2004
Computer Systems & Architectures	3rd year Computer science		9/2002-1/2003
Artificial Intelligence	3rd year	Faculty of Computer & Information Sciences Ain shams University	2/2004□ 6/2004
Operating Systems	3rd year		2/2008
Compiler Theory	4th year Computer science		9/2002-1/2008
Computer Security	4th year Computer science		2/2002-6/2002 2/2003-6/2003
Digital Design	2nd year		9/2001-1/2002
Advanced Compilers Design	Post-graduate Computer Science		10/2003-2/2004 10/2002-2/2003
Introduction to wireless networks	4th year Computer science		1/2005 -5/2005
Parallel & distributed systems	4th year Computer science		1/2005 -5/2005
Structured Programming	1st year		1/2006- 5/2008
Distributed Operating Systems	Post-graduate computer science		10/2005-2/2006
Report Writing			2/2006-6/2006
Preparing Research Material			
Computing Algorithms	10th &7th term Computer Engineering	Arab Academy for Science & Technology	2/2003-6/2003
Advanced Programming	3rd term Computer science		
Digital Design	3rd term-Computer Science & Computer Engineering		
Programming Applications	Summer course		7/2003
Computer Systems	4th year Electrical Engineering	Alexandria Institute of Technology	9/2003-1/2004
Algorithms Design & Analysis	2nd year Computer Engineering		
Information Technology	3rd year Computer Engineering		2/2004-6/2004
Formal Languages	3rd year Computer Engineering		
Operating Systems	3rd year Computer Engineering	Pharos University	9/2009-2/2010
Computer & Networks Security	4th year Computer Engineering	Pharos University	3/2011-7/2011
Data Structures	2nd year Computer Engineering	Beirut Arab University	2/2012-4/2012

Reference Professors

Prof. Dr. Hesham El-Rewini:
Interim Provost, Oregon Institute of Technology, hesham.elrewini@oit.edu
(was a PhD examiner & foreigner counterpart in a post doctoral)
https://github.com/DrShymaa2022/recommendation_letters_CVdocuments/blob/main/Prof_ElRewini_Marymount_VicePresident.pdf

-Prof. Dr. Nagwa El-Makky:
Professor in Alexandria University, nagwamakky@alexu.edu.eg
(was a teaching Professor, a master supervisor, and attended my PhD defense)
https://github.com/DrShymaa2022/recommendation_letters_CVdocuments/blob/main/DrNagwa_correct_ref_let.pdf

-A third reference name to contact if needed:
Prof. Dr. Essam Khalifa
Vice President/consultant, Egyptian Chinese University, Esskhalifa49@gmail.com
(was the last Dean in my last Tenure Track Position at Ain Shams University)

Masters I've Supervised

Intelligent Technique For Computer Virus Detection (**papers 8 & 13**)
https://srv3.eulc.edu.eg/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=PublicDrawThesis&BibID=11104058

Operating Systems Scheduling Algorithms (**papers 9 & 11**)
https://www.eulc.edu.eg/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=PublicDrawThesis&BibID=9581512

Study Of Encryption Algorithms (**papers 14 & 16**)
https://srv3.eulc.edu.eg/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=PublicDrawThesis&BibID=11107902

A secure routing protocol for mobile AD-Hoc networks (**paper 15**)
https://srv3.eulc.edu.eg/Eulc_v5/Libraries/start.aspx?fn=ApplySearch&ScopeID=&SearchText1=A+secure+routing+protocal+for+mobile+AD-Hoc+networks&criteria1=0.
(My name does not appear because only the main supervisor is listed and the thesis was in another faculty, Engineering, than the one I was working in. However, you can see from the paper in 15 that I've participated in the thesis; the thesis resulted in one paper only, but it was a stronger paper that was accepted in IEEE and was eventually published in a journal).

A Machine Learning Approach for Web Usage Mining
https://srv3.eulc.edu.eg/eulc_v5/Libraries/Thesis/BrowseThesisPages.aspx?fn=PublicDrawThesis&BibID=11112306
(I was not closely involved with the scientific contribution of this thesis; I've only participated in the writing guidance and review process. In fact, I joined the supervision to solve a procedural problem and resigned before the final debate)

Publications

1- S. Arafat, A. Belal, N. el-Mekky, "A new parallel sample sorting algorithm", *In Proceedings of the IASTED conference on Computer Systems & Applications*, 30 March- 2 April 1998, Irbid, Jordan.

2- A.Belal, M. Abd el Azim, S. Arafat, "Limiting the search for two dimensional optimal alphabetic trees", *In Proceedings of The Fourth Joint Conference for Information Sciences JCIS98*, 23-28 October 1998, North Carolina, USA; http://dimacs.rutgers.edu/archive/TechnicalReports/abstracts/1998/98-43.html, full paper: https://github.com/DrShymaa2022/PhD_papers/blob/main/LimitingSearchFor2DoptimalAlphabeticTrees_98.pdf

3- A. Belal, M. Selim, S. Arafat, "Rebuilding Huffman Trees in Linear Time", *In Proceedings of The Seventh Maghrebian Conference on Computer Sciences MCSSEAI'2002*, 6-8 May 2002, Annaba, Algeria.

4- A. Belal, M. Selim, S. Arafat, "Merging Optimal Alphabetic Trees in Linear Time", *In Proceedings of The First International Conference on Intelligent Computing & Information Systems ICICIS 2002*, 24-16 June 2002, Cairo, Egypt.

5- A. Belal, M. Selim, S. Arafat, "Building Optimal Alphabetic Trees Recursively", *The 3rd WSEAS International Multiconference on Mathematics & Math 2002*, 1-5 September 2002, Wolin Island, Poland, Vol 1, Issue2 of WSEAS Transaction on Mathematics, 2002, pp.77-82.

6- A. Belal, M. Selim, S. Arafat, "*Towards an Optimal Dynamic Alphabetic Tree*", International journal on Intelligent Computing & Information Systems IJICIS; http://archive.dimacs.rutgers.edu/TechnicalReports/abstracts/1999.sav/99-22.html, full paper: https://github.com/DrShymaa2022/PhD_papers/blob/main/TowardsDynamicOptimalAlphabeticTree.pdf

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After PhD

7- S. Arafat, "An Efficient simple Algorithm for Building Optimal Alphabetic Trees in Parallel", *WSEAS International Conference on Algorithms and Scientific Computing,* Vol 2, Issue3 of WSEAS Transaction on Computers, July 2003.

8- M.Syam, M. Hashem, S. Arafat, M .Mabrouk, "Neural Networks for Computer Virus Detection", *The 4th WSEAS International Conference on Neural Networks and Applications,* May 2003, Athens, Greece, Vol 2, Issue3 of WSEAS Transaction on Systems, July 2003.

9- M.Syam, Z. Fayed, S. Arafat, N.Hossam el Din, "Efficient Probabilistic Fault-tolerant Scheduling Algorithm for Real-time Tasks in Heterogeneous Multi-Processor Systems", *The 4th WSEAS International Conference on Computer Systems*, Dec 2004, *New York.* Vol 2, Issue3 of WSEAS Transaction on Systems, April 2004.

10- S. Arafat, "Encryption Using Alphabetic Trees", *The 3rd ACS/IEEE International Conference on Computer Systems and Applications, (AICCSA'05)*, Jan 2005, Cairo, Egypt; https://ieeexplore.ieee.org/abstract/document/1387083/

11- M.Syam, Z. Fayed, S. Arafat, N. .Hossam el Din, "An Efficient Online Scheduling Algorithm For Real Time Tasks In Multi Processor Systems", *2nd International conference on Intelligent computing & Information Systems ICICIS*, 5-7 March 2005, Cairo, Egypt.

12- M.Syam, S.Arafat, A.Ibrahim, "A CFG Based Authorization Frame Work", *2nd International conference on Intelligent computing & Information Systems ICICIS*, 5-7 March 2005, Cairo, Egypt. (my role in this paper was minor, solely the student idea)

13- M.Syam, M. Hashem, S. Arafat, M .Mabrouk, "Data Mining for Computer Virus Detection", *2nd International conference on Intelligent computing & Information Systems ICICIS*, 5-7 March 2005, Cairo, Egypt.

14- M. Tolba, S. Arafat, R. Kotb, "Analysis of Alphabetic Tree Cipher", *the 4th International Multi conference in Computer Science & Information Technology*, 5-7 April 2006, Amman, Jordan.

15- S.Arafat, Y.Dakroury, A.Wahab, *"An Efficient and Secure Ad hoc Routing Protocol"*, Global Engineering Science and Technology Society (GESTS) Transaction On Computer Science & Engineering, Vol. 32 and No.1, Sept.2006; https://dblp.org/pid/81/7056.html (was accepted in IEEE conference)

16- M. Tolba, S. Arafat, R. Kotb, *"Comparing Alphabetic Tree Cipher (ATC) to RC5, RC6 and Rijndael"*, International journal on Intelligent Computing & Information Systems IJICIS, 2006.

Back From Retirement
(in chronological order)

17- Draft talk on Algorithmic Game Theory (12/2019)
https://m.facebook.com/story.php?story_fbid=1018702451817536&id=100010333725264

18- Draft talk on Blockchain Market Manipulation Attacks (3/2021)
https://m.facebook.com/story.php?story_fbid=1386287771725667&id=100010333725264

19- S. Arafat, "*Partitioning Bitcoin UTXOS Merkle Tree Into Different SubTrees To Reduce Proof Size*", https://github.com/Shymaa-Arafat/lab2/blob/main/DeFi_2UTXOS2_NewOutline.pdf; (rejected by DeFi'22) Reviewer1: border line paper, reviewers2&3: Bitcoin issue, not related to DeFi. The detailed review reports can be found on the Facebook link.
https://m.facebook.com/story.php?story_fbid=1567082160312893&id=100010333725264

20- S. Arafat, "*Providing Non-inclusion Proofs in Bitcoin append to right UTXOS Merkles"*, https://github.com/Shymaa-Arafat/lab2/blob/main/DeFi_3fraudProofs.pdf; rejected by DeFi'22, Reviewer1&2: Bitcoin issue, not related to DeFi. The detailed review reports: https://m.facebook.com/story.php?story_fbid=1566310467056729&id=100010333725264

21- S. Arafat, " *The Effect of Propagation Delay Due to Geo-distribution on Blockchains*", https://github.com/Shymaa-Arafat/lab2/blob/main/DeFi_MOOC_Article.md, submitted in partial fulfillment of the DeFi MOOC course 2021

22- S. Arafat, " *SoK : Can NFTs Solve The Economic Problems of Countries with Ancient Heritage? Egypt as a Case Study"*, https://github.com/Shymaa-Arafat/lab2/blob/main/CameraReady_1sprngr_EgyptN.pdf, https://youtu.be/mT7qSMvV7FQ; **accepted** by EAI BlockTEA 2022; **published** by Springer Verlag: https://link.springer.com/chapter/10.1007/978-3-031-31420-9_10.

23- S.Arafat, https://medium.com/@shymaa.arafat/what-are-verkle-trees-kzg-commitments-and-could-they-be-applied-on-bitcoin-cbf4838d18ac; an article submitted (on 6 May 2023) in partial fulfilment of the *Trailerblaze*r Tier of the ZK MOOC course 2023.

24- S. Arafat, https://medium.com/@shymaa.arafat/fiat-shamir-transformation-and-its-security-problems-shymaa-m-a14d8f7d9192; an article submitted (on 6 May 2023) in partial fulfilment of the *Trailerblazer* Tier of the ZK MOOC course 2023.

25- S. Arafat, https://medium.com/@shymaa.arafat/metaverse-threats-liabilities-c7d8fd50a21d; 18/6/2023.

26- S. Arafat, https://medium.com/@shymaa.arafat/defi-attacks-part-1-1906a9307fe2; 21/6/2023.

27- S. Arafat, https://medium.com/@shymaa.arafat/defi-attacks-part-2-45c7e2c9c00; 25/6/2023.

28- S. Arafat, https://medium.com/@shymaa.arafat/cbdcs-part1-1da3a944722c; 7/2023

29- S. Arafat, https://medium.com/@shymaa.arafat/what-happened-to-polynetwork-2021-then-2023-2e991e29655b; 23/10/2023.

30- S. Arafat, https://github.com/DrShymaa2022/articles_papers/blob/main/Understanding_CrossChain_Interoperability_Solutions-_part1.pdf; (available on medium too)30/10/2023

31- S. Arafat, https://medium.com/@shymaa.arafat/bugs-in-our-pockets-client-side-scanning-from-2021-to-2024-2442b286e501, presentation: https://youtu.be/Qq3FT3CferU

32- S.Arafat, https://medium.com/@shymaa.arafat/crypto-crimes-2023-2024-report-b73055ba9d8b

33- S. Arafat, "*On the Estonian Internet Voting System, IVXV, SoK and Suggestions*", https://www.researchgate.net/publication/383220628_On_the_Estonian_Internet_Voting_System_IVXV_SoK_and_Suggestions; extended version: https://eprint.iacr.org/2025/506

34- S. Arafat, "*Blockchains in Water Management & Trading*", https://www.researchgate.net/publication/382642184_A_Study_of_Blockchain_applications_in_the_Water_Sector; a preprint.

35- S. Arafat, "*A Study of Blockchain Consensus Protocols*", https://www.researchgate.net/publication/382447691_A_Study_of_Blockchain_Consensus_Protocols; https://eprint.iacr.org/2025/637; this version is **accepted** by Cyber Security Journal (http://www.cybersecurity.springer.com/); DOI: 10.1186/s42400-025-00417-x, **to be printed** in 2026.

36- S. Arafat, "WIP: *Towards Hardcopy Solid Court Verifiability- A (2 Devices) (3 Receipts) in-Booth Electronic Voting System*"; https://github.com/DrShymaa2022/articles_papers/blob/main/courtverify_EVoteID_final.pdf; earlier presentation: https://youtu.be/0wzcHdXEKHM

37- The main idea of [36] was **published as a poster** (based on abstract review only) in E-Vote-Id 2025 (https://e-vote-id-2025.inria.fr/pages/program.html); Poster: https://github.com/DrShymaa2022/E_Vote_ID_2025_posters/blob/main/CourtVerify2_P_Poster_9_9_25.pdf; Teaser video: https://youtu.be/7q7c7p8HqKI

38- S. Arafat, "*Automated Ballot Stuffing with an Encrypted Vote: A Large-Scale Attack on the Estonian Internet Voting system, IVXV, and its Mitigation*", **published** as E-Vote-Id 2025 **poster** (based on abstract review only); Poster: https://github.com/DrShymaa2022/E_Vote_ID_2025_posters/blob/main/EncryptedcopyattackPoster_13_9_25.pdf; Teaser video: https://youtu.be/QRmtfowimOg

39- S. Arafat, "*Removing Insiders' Trust from the Estonian Internet Voting System (IVXV)*", **published** as E-Vote-Id 2025 **poster** (based on abstract review only); Poster: https://github.com/DrShymaa2022/E_Vote_ID_2025_posters/blob/main/Insiders2TrustPoster_7_9_25.pdf; Teaser video: https://youtu.be/joQ12G6KGzc